



# Guideline-based and restricted fluid resuscitation strategy in sepsis patients with heart failure: A systematic review and meta-analysis

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## ABSTRACT

**Objectives:** To examine whether a fluid resuscitation strategy based on guidelines (at least 30 mL/kg IV crystalloids) vs. a restrictive approach with <30 mL/kg within three hours affects in-hospital mortality in patients with sepsis and a history of heart failure (HF).

**Data sources:** On 03/07/2023, we searched Embase, PubMed, and Scopus for peer-reviewed papers and abstracts using the PRISMA guidelines.

**Study selection:** The language was limited to English. Studies published since 2016 included if they had sepsis patients with a history of HF, or a subgroup of patients with HF, and in-hospital mortality data on these patients that did or did not meet the 30 mL/kg by 3 h (30 × 3) goal. Duplicate studies, studies that focused on a broader period than 3 h from the diagnosis of sepsis or without mortality breakdown for HF patients or with unrelated title/abstract, or without an IRB approval were excluded.

**Data extraction:** In-hospital mortality data was taken from the final studies for HF patients with sepsis who did or did not meet the 30 × 3 goal.

**Data synthesis:** The meta-analysis was performed using the Review Manager 5.4 program with ORs as the effect measure. The ProMeta program version 3.0 was used to evaluate the publication bias. Egger's linear regression and Berg and Mazumdar's rank correlation was used to evaluate the publication bias. The result was visually represented by a funnel plot. To estimate the proportion of variance attributable to heterogeneity, the  $I^2$  statistic was calculated.

**Results:** The search yielded 26,069 records, which were narrowed down to 4 studies. Compared to those who met the 30 × 3 goal, the <30 × 3 group had a significantly higher risk of in-hospital mortality (OR = 1.81, 95% CI = 1.13–2.89,  $P = 0.01$ ).

**Conclusions:** Restrictive fluid resuscitation increased the risk of in-hospital mortality in HF patients with sepsis. More rigorous research is required to determine the optimal fluid resuscitation strategy for this population.

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## 1. Introduction

Sepsis is associated with significant burden and cost to the health care system in the United States. A study among Medicare beneficiaries from 2012 to 2018 showed an increase in hospital admissions for sepsis from 811,644 to 1,136,889, with total cost increase from \$17,792,657,303 to \$22,439,794,212 [1]. In the same study, 6-month mortality was 60% for septic shock, 36% for severe sepsis, 31% for sepsis attributed to a specific organism, and 27% for unspecified sepsis.

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The Third International Consensus Definitions for Sepsis and Septic Shock defines sepsis as life-threatening organ dysfunction caused by a dysregulated host response to infection, with organ dysfunction being represented by a change in Sequential Organ Failure Assessment (SOFA) score by 2 or more from baseline [2].

The most recent 2018 Surviving Sepsis Campaign Bundle (SSCB) update considered intravenous (IV) fluid resuscitation as the cornerstone of sepsis treatment. This expert consensus statement recommended administration of at least 30 mL/kg of IV crystalloids within the first 3 h of recognition of sepsis when hypotension and/or elevated lactate is present without exceptions for patients with preexisting comorbidities that may be affected by volume overload, including heart failure (HF) [3]. However, when it comes to volume resuscitation, sepsis and concomitant HF provide a particularly

difficult clinical scenario. Several publications reported a link between positive fluid balance in septic patients and mortality [4–6]. On the other hand, adherence to a minimum 30 mL/kg bolus was not linked to increased mortality in a multicenter study comparing treatment before and after sepsis bundle implementation [4]. One multicenter observational study found that sepsis bundle implementation was associated with improved volume resuscitation and increased survival in patients with sepsis and concomitant HF [5].

Recently the CLOVERS trial concluded that with sepsis-induced hypotension, the restrictive fluid strategy did not increase (or decrease) mortality or safety concerns before discharge on day 90 compared to the liberal fluid strategy [6]. These and other findings explain why some physicians are hesitant about ordering fluids for septic patients at higher risk of volume overload. The ideal resuscitation strategy is unknown, and whether early guideline-based volume resuscitation is appropriate for individuals with HF is controversial. One study showed that crystalloid was initiated significantly later in patients with HF or renal failure [7]. Other studies reported that patients having a prior medical history of HF were more than twice as likely to receive less fluid [8].

Although fluid resuscitation trials on patients with sepsis and a history of HF stratified by severity are crucial to addressing fluid resuscitation goals in these patients, observational studies provide some insight on the impact of the guidelines in practice. Due to inconsistent and insufficient evidence, we conducted a systematic review and meta-analysis within the available literature to shed some light on this dilemma.

We examined whether a restricted fluid resuscitation strategy of  $<30$  mL/kg IV crystalloids within the first three hours ( $<30 \times 3$ ) affects in-hospital mortality compared to the SSCB recommendation of at least 30 mL/kg by 3 h ( $30 \times 3$ ) in sepsis patients with a history of HF.

## 2. Materials and methods

### 2.1. Hypothesis

We hypothesized that a restricted fluid resuscitation strategy using  $<30 \times 3$  IV crystalloids, would not significantly affect in-hospital mortality, compared to adherence to the  $30 \times 3$  SSCB recommendation in patients with a history of HF.

### 2.2. Method of search

A systematic review and meta-analysis was conducted according to PRISMA guidelines. A search in Embase, PubMed, and Scopus databases was performed on 03/07/2023. The following search syntax was used to search peer-reviewed papers, abstracts, and RCTs, based on their title, abstract, and keywords: “heart failure” or “congestive heart failure” or “heart decompensation” or “cardiac failure” and “sepsis” or “septic shock” and “fluid resuscitation” or “crystalloid” or “30 by 3”, and “mortality” or “in-hospital mortality” (Supplemental table 1).

### 2.3. Selection criteria

The language was limited to English. One author (AW.) reviewed the search result titles and abstract, and two authors (AW., AV.) independently reviewed the eligible records. Disagreements over the selection of relevant papers were resolved through discussion and consensus. Studies published since 2016 included if they had sepsis patients with a history of HF (or a subgroup of patients with HF) and in-hospital mortality data on the patients that did or did not meet the  $30 \times 3$  goal. Duplicated records, studies with unrelated title/abstract, studies that focused on a period broader than 3 h from the diagnosis of sepsis, and studies without mortality breakdown for HF patients or without IRB approval were excluded.

### 2.4. Search result

The search yielded 26,069 records, which were reduced to 12,549 after removing duplicates. After removing records with irrelevant titles or abstracts, this decreased to 9 (Fig. 1). Of these, 2 were excluded because no clear timeframe for fluid administration was provided, and 3 others were excluded because no raw mortality data was provided. After assessment for risk of bias by three authors, 4 final studies were considered for analysis (Table 1).

### 2.5. Assessing risk of bias

The Newcastle-Ottawa Scale for case-control studies and retrospective cohorts was used by three authors (AW., AV., AC.) to assess the risk of bias in the included studies (Supplemental Table 2,3,4). The included studies received a rating of  $>6$  stars by each author.

### 2.6. Statistical analyses

The meta-analysis was performed using the Review Manager 5.4 program (The Cochrane Collaboration, 2020) [9] by the random effects model with OR as the effect measure. The OR and 95% CI for mortality among HF patients with sepsis who received  $<30 \times 3$  vs.  $30 \times 3$  were calculated for each study. The mortality data were totaled, and the studies were weighted to calculate a total OR and 95% CI (Fig. 2). The program calculated the  $I^2$  statistic for heterogeneity analysis.  $I^2$  estimates the proportion of variance that is due to heterogeneity. This metric denotes the proportion of variation throughout the included studies that is due to heterogeneity rather than chance [10,11]. The ProMeta program version 3.0 [12] was used to evaluate publication bias. This program generated a funnel plot (Fig. 3), and determined the degree of asymmetry using Egger's linear regression test and Begg and Mazumdar's rank correlation test (Fig. 4). [13] [14].

## 3. Results

### 3.1. Characteristics of the studies included in the meta-analysis

Table 1 shows an overview of the studies used for our meta-analysis. Kuttub et al. [15] found that a history of HF was associated with decreased odds of  $30 \times 3$  (OR = 0.42; 95% CI = 0.29–0.60); this retrospective cohort study showed that failure to reach  $30 \times 3$  was associated with increased odds of mortality (OR = 1.52; 95% CI = 1.03–2.24), delayed hypotension (OR = 1.42; 95% CI = 1.02–1.99), and increased intensive care unit (ICU) stay ( $\sim 2$  d) ( $\beta = 2.0$ ; 95% CI = 0.5–3.6) for patients with sepsis, without differential effects for “at-risk” groups including those with obesity, end-stage renal disease (ESRD), HF, or advanced age. One limitation mentioned is that the group with adequate fluid resuscitation had inherent differences compared with the other group, such as higher prevalence of septic shock, reduced time to antibiotics by  $\sim 1$  h, and likely increased recognition of sepsis with slightly higher rates of Systemic Inflammatory Response Syndrome and Quick SOFA scores on presentation.

Uchel et al. [16] showed that although administration of  $30 \times 3$  in HF patients with sepsis resulted in 2.5 times more intubations (23% vs. 9%;  $p < 0.001$ ), and 2.15 times more readmissions at 30 days (28% vs. 13%;  $p < 0.001$ ), HF patients who received  $<30 \times 3$  had doubled the mortality (34% vs. 17%;  $p = 0.0027$ ).

Taenzer et al. [17] used Medicare claims data to conduct a retrospective cohort study on the survival of elderly patients with severe sepsis or septic shock and a history of HF who did or did not reach  $30 \times 3$  upon presenting to the emergency department. They used a proportional Cox hazard model to compare the two groups' 1-year survival rates and found that patients who received  $<30 \times 3$  had higher adjusted in-hospital and post-discharge mortality rates (hazard ratio = 2.18; 95% CI = 1.2–4.0;  $p = 0.01$ ). They concluded that the administration of

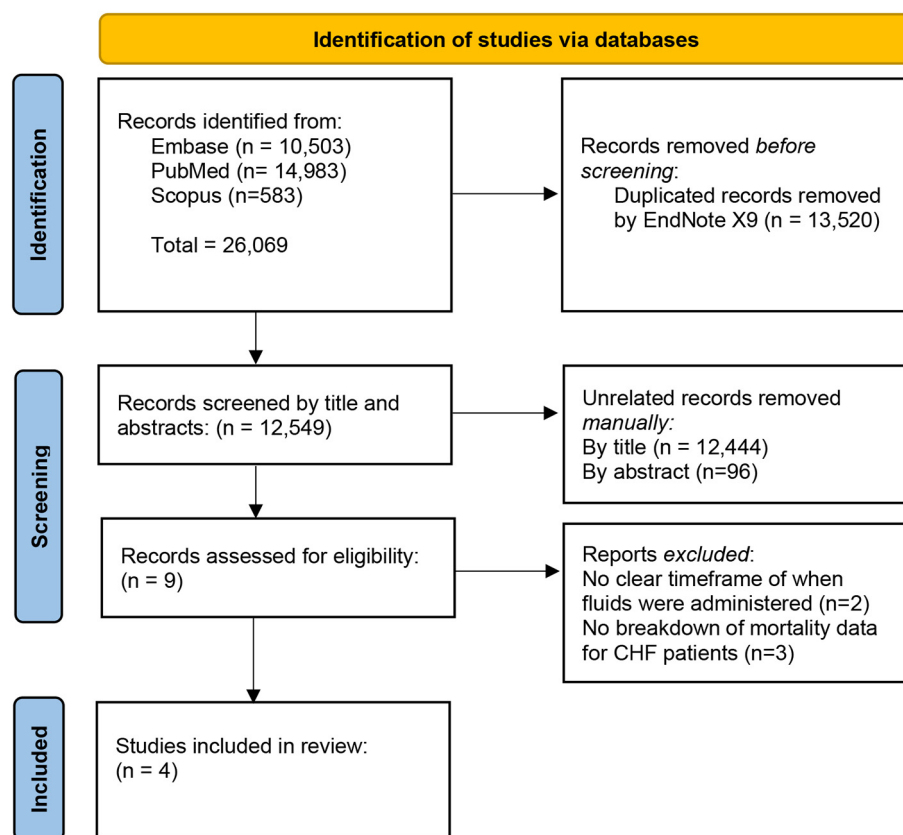


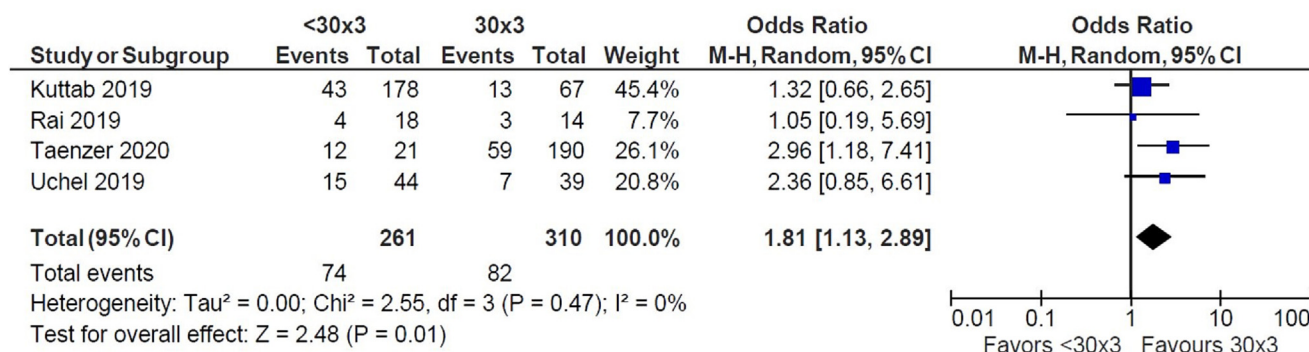
Fig. 1. Search result.

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;372:n71. doi: 10.1136/bmj.n71. For more information, visit: <http://www.prisma-statement.org>

**Table 1**  
Overview of the studies included in the meta-analysis.

| Study               | Publication Type/Study Design                        | Type of Fluid Used | Overall Baseline Characteristics                                                                                                                                                                                                                                           | Characteristics of HF <sup>§</sup> Patients with Sepsis who Received <30 × 3                                                                                                                                                                                                                                  | Characteristics of HF <sup>§</sup> Patients with Sepsis who Received 30 × 3                                                                                                                                                                                                                    | Findings                                                                                                                                                                                           |
|---------------------|------------------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kuttab et al. [15]  | Original research article/Retrospective cohort study | Crystalloid        | ED <sup>†</sup> treated adults, predominantly serving African American patients. Controlled for sex, ESRD <sup>‡</sup> , HF <sup>§</sup> , sepsis severity (severe sepsis vs septic shock), obesity, mortality in ED <sup>†</sup> , Sepsis score, and time to antibiotics. | Not described for this subgroup                                                                                                                                                                                                                                                                               | Not described for this subgroup                                                                                                                                                                                                                                                                | Failure to meet 30by3 increased odds of mortality (OR <sup>¶</sup> , 1.52; CI*, 1.03–2.24)                                                                                                         |
| Rai et al. [18]     | Abstract/Retrospective cohort study                  | Not mentioned      | Sepsis patients, patients with history of HF <sup>§</sup> adjusting for age, sex, race, and Charlson score.                                                                                                                                                                | Not described for this subgroup                                                                                                                                                                                                                                                                               | Not described for this subgroup                                                                                                                                                                                                                                                                | Mortality was similar in those who received >30 mL/kg fluids within 3 h (21.4%) vs those who did not (22.2%, <i>p</i> = 0.957; adjusted OR <sup>¶</sup> 0.91, 95% CI* 0.16–5.12, <i>p</i> = 0.914) |
| Uchel et al. [16]   | Abstract/Retrospective cohort study                  | Not mentioned      | Patients with a diagnosis of sepsis or septic shock with underlying compensated HF <sup>§</sup> , 33 systolic, 48 diastolic, 2 combined. No hemodialysis dependents or with liver cirrhosis patients were included.                                                        | Systolic HF <sup>§</sup> : 19/44 (43.2%)<br>Diastolic HF <sup>§</sup> : 24/44 (54.5%)<br>Combined HF <sup>§</sup> : 1/44 (2.3%)<br>Age (mean ± SD <sup>§</sup> ): 82 ± 10.56<br>Male: 1/39 (2.6%)<br>Septic shock: 11/21 (52%)<br>Hierarchical condition category score (mean ± SD <sup>§</sup> ): 3.4 ± 2.46 | Systolic HF <sup>§</sup> : 14/39 (35.9%)<br>Diastolic HF <sup>§</sup> : 24/39 (61.5%)<br>Combined: 1/39 (2.6%)<br>Age (mean ± SD <sup>§</sup> ): 83 ± 10.0<br>Male: 93/190 (49%)<br>Septic shock: 101/190 (53%)<br>Hierarchical condition category score (mean ± SD <sup>§</sup> ): 4.4 ± 2.31 | Mortality was 17% in the liberal group and 34% in the restrictive group ( <i>p</i> value 0.0027).                                                                                                  |
| Taenzer et al. [17] | Original research /Retrospective cohort study        | Crystalloid        | Sepsis patients, 65 years and older with a history of HF <sup>§</sup> admitted through ED <sup>†</sup> , ejection fraction <35%.                                                                                                                                           |                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                | The restrictive group had higher adjusted in-hospital (57% vs 31%; <i>P</i> = 0.01) and post-discharge mortality rates.                                                                            |

\* CI=Confidence Interval, <sup>†</sup>ED = Emergency department, <sup>‡</sup>ESRD = End-Stage Renal Disease; <sup>§</sup>HF=Heart Failure; <sup>¶</sup>OR = Odds Ratio; <sup>§</sup>SD=Standard Deviation.



**Fig. 2.** Forest plot displaying the weighted odds ratios with 95% CI of in-hospital mortality in the <30×3 vs. the 30×3 group from all included studies. The overall summary effect is represented as a diamond shape.

30 × 3 is associated with lower mortality. Aside from the retrospective nature of their study, the number of patients analyzed and the small cohort of patients who received <30 × 3 limited their findings.

Using Centers for Medicare and Medicaid Services data, Rai et al. [18], analyzed the association between receiving greater than versus <30 mL/kg fluid and in-hospital mortality (adjusting for age, sex, race, and Charlson score) in a subset of patients with a history of HF and sepsis. They found that the mortality was similar in those who received 30 × 3 vs. those who did not ( $OR = 0.91$ , 95% CI = 0.16–5.12,  $p = 0.914$ ). They suggested that a larger cohort is required to support the benefit of 30 × 3 recommendation and its safety in patients with HF.

### 3.2. Statistical analysis results

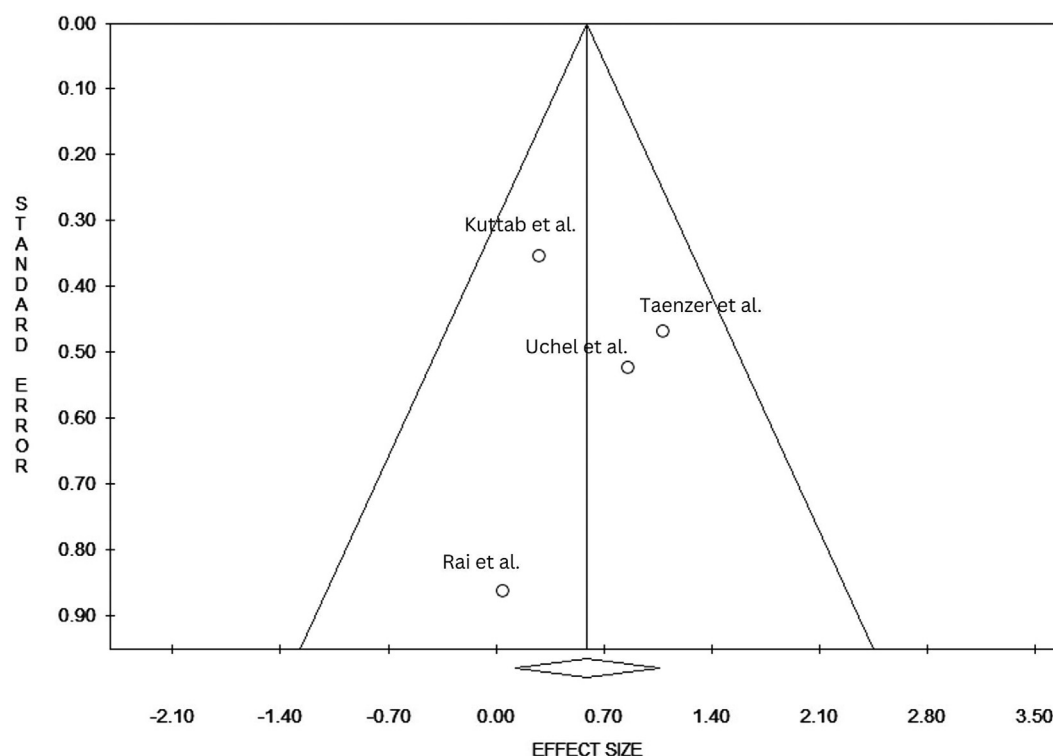
When the data from the included studies were pooled, 571 patients with sepsis and a history of HF were identified. In the <30 × 3 group, 74 of the 261 individuals died, and the number of deaths in the 30 × 3

group was 82 of the 310 individuals (Table 2). The weighted OR for death was significantly higher in the <30 × 3 vs. 30 × 3 ( $OR = 1.81$ , 95% CI = 1.13–2.89,  $p = 0.01$ ). Heterogeneity analysis resulted in an  $I^2$  of 0% (Fig. 2), demonstrating a high degree of homogeneity among the results of the selected studies and pooled data.

### 3.3. Risk of bias

The Newcastle-Ottawa Scale for case-control studies and retrospective cohorts was used by three authors (AW., AV., AC.) to assess the risk of bias in the included studies, all of which received a rating of >6 stars by each author (Supplemental Table 2,3,4).

Publication bias analysis was performed by Egger's linear regression and Begg and Mazumdar's rank correlation test and displayed by funnel plot (Fig. 3). Egger's test revealed a  $p$  value of 0.95; and Begg and Mazumdar's rank test revealed a  $p$  value of 1.0, both suggesting no significant publication bias. This is visually demonstrated by a symmetric funnel plot.



**Fig. 3.** Funnel plot illustrating symmetry within the included studies.

## Publication bias analysis

Trim and fill analysis (random-effects model)

Plot

| Id                              | ES   | LL   | UL   | Sig.  | V | SE |
|---------------------------------|------|------|------|-------|---|----|
| Overall effect size (observed)  | 1.81 | 1.13 | 2.89 | 0.013 |   |    |
| Overall effect size (estimated) | 1.81 | 1.13 | 2.89 | 0.013 |   |    |

\* Trimmed studies: 0

Egger's linear regression test

Intercept 0.14 t 0.07 Sig. 0.950

Begg and Mazumdar's rank correlation test

Z value for Kendall's tau 0.00 Sig. 1.000

Fail Safe N 2 The value is below Rosenthal's rule of thumb ( $5k+10 = 30$ )

Fig. 4. Results of the publication bias analysis using Egger's linear regression test and Begg and Mazumdar's rank correlation test.

## 4. Discussion

Our meta-analysis finds a significantly increased risk of in-hospital mortality among hospitalized HF patients with sepsis/septic shock who received a restricted resuscitation strategy compared to those who met the  $30 \times 3$  goal. This finding is consistent with that of Taenzer et al. and is aligned with the findings of other included research that, while not statistically significant, pointed in the same direction (Fig. 2).

Other studies like Wiczorek et al. [19] found no statistically significant association between resuscitation with  $>30 \times 3$  in their at-risk group (HF and/or chronic kidney disease patients) with severe sepsis/septic shock and in-hospital mortality by 3 or 6 h. Likewise, Chiang et al. [20] found no significant difference in mortality in patients with HF or ESRD and sepsis/septic shock who received  $<30$  mL/kg by 3 h. However, the  $>30 \times 3$  group had lower mortality at 6 h. The breakdown of mortality data in these two studies for the  $30 \times 3$  group was not provided, so we were unable to incorporate them into our meta-analysis. Acharya et al. [21] found that

administration of at least 30 mL/kg IV fluids in the first 6 h was associated with decreased odds of mortality in HF patients with severe sepsis/septic shock. Their data were excluded from our meta-analysis because they focused on bolus delivery within the first 6 h following sepsis diagnosis instead of 3.

We considered analyzing a critical challenge in implementing SSCB-recommended fluid resuscitation in a subgroup of patients who cannot be randomly assigned to one or other treatment regimens due to ethical constraints. The method we used was based on a validated systematic review method and clinical factors. We used three different databases to capture a large potential pool of sources using the same search terms. However, only 4 out of 12,549 unique records with mortality data for HF patients who did or did not meet the  $30 \times 3$  fluid resuscitation recommendation were identified. This database search result highlighted the scarcity of comparable studies on this topic. All 4 included studies discussing the administration of  $30 \times 3$  to HF patients with sepsis were retrospective and observational, therefore a risk of confounding and data limitation must be considered. In addition, 3 of the 4 studies were abstracts.

Table 2

Mortality rates in  $<30 \times 3$  and  $30 \times 3$  groups based on raw data extracted from the individual selected studies.

| Study                     | Number of patients with sepsis and history of HF | Number of patients on restricted Fluid resuscitation Strategy ( $<30 \times 3$ ) | Number of deaths in restricted Fluid resuscitation Strategy ( $<30 \times 3$ ) | Number of patients on guideline-based strategy ( $30 \times 3$ ) | Number of deaths in guideline-based strategy liberal strategy ( $30 \times 3$ ) |
|---------------------------|--------------------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Kuttab et al. [15], 2019  | 245                                              | 178 (73%)                                                                        | 43 (24%)                                                                       | 67 (27%)                                                         | 13 (19%)                                                                        |
| Rai et al. [18], 2019     | 32                                               | 18 (56%)                                                                         | 4 (22%)                                                                        | 14 (44%)                                                         | 3 (21%)                                                                         |
| Uchel et al. [16], 2019   | 83                                               | 44 (53%)                                                                         | 15 (34%)                                                                       | 39 (47%)                                                         | 7 (18%)                                                                         |
| Taenzer et al. [17], 2020 | 211                                              | 21 (10%)                                                                         | 12 (57%)                                                                       | 190 (90%)                                                        | 59 (31%)                                                                        |
| Total                     | 571                                              | 261 (46%)                                                                        | 74 (28%)                                                                       | 310 (54%)                                                        | 82 (26%)                                                                        |

Selection bias, publication bias and other confounders are inevitable in this type of research. We sought to overcome these and other methodological concerns by following the PRISMA guidelines, carefully and restrictively specifying the inclusion/exclusion criteria, and employing a clear definition of the exposure and outcome. Using these measures in this study, however, influences the number of selected studies and the sample size, thereby reducing its statistical power. In conducting our analysis, we used several methods to reduce potential bias due to study selection. Our formal literature search was performed without limiting the search by the outcome. All studies included were retrospective or observational, obviating the need to search prospective trial registers. The studies were reviewed by three authors independently. Publication bias analysis was performed by Egger's linear regression; and Begg and Mazumdar's rank correlation tests; and displayed by funnel plot, demonstrating low likelihood of publication bias. However, the point estimate  $I^2$  should be interpreted with caution when there are few studies in a meta-analysis [22].

## 5. Conclusion

There is scant evidence to guide acute resuscitation of high-risk septic patients, such as those with a history of HF, and providers' adherence to sepsis bundles in HF patients remains variable. Earlier trials (Rivers, ProCESS, ARISE, and ProMISe) analyzed IV fluid as part of a comprehensive sepsis protocol. Therefore, assumptions based solely on IV fluid approaches in high-risk patients are limited. Confounders, on the other hand, would impact observational research based on retrospective data. There is a clinical debate about the optimal IV fluid resuscitation strategy. We merely evaluated the existing evidence by a meta-analysis approach. Based on the published research, our meta-analysis revealed that a restricted fluid resuscitation strategy (<30 mL/kg fluids within 3 h) in sepsis/septic shock patients with a history of HF was associated with increased risk of in-hospital mortality, while the included primary studies didn't achieve a statistical significance, except for the Taenzer et al. study.

Our study highlights the need for extensive research to properly address the fluid resuscitation approach in this group of patients. Additional research in larger and more diverse populations may also consider the impact of HF with reduced ejection fraction and preserved ejection fraction phenotypes, the impact of pressors, the severity of sepsis, non-ICU sepsis admissions, and use of invasive hemodynamic monitoring to guide resuscitation.

## CRediT authorship contribution statement

**Ali Vaeli Zadeh:** Writing – review & editing, Writing – original draft, Supervision, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Alan Wong:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Andrew Carl Crawford:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Data curation, Conceptualization. **Elias Collado:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Joshua M. Larned:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

## Declaration of Competing Interest

None.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ajem.2023.08.006>.

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